Inhibition of bacterial adherence by cranberry juice: potential use for the treatment of urinary tract infections.

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Cranberry juice has been widely used for the treatment and prevention of urinary tract infections and is reputed to give symptomatic relief from these infections. Attempts to account for the potential benefit derived from the juice have focused on urine acidification and bacteriostasis. In this investigation it is demonstrated that cranberry juice is a potent inhibitor of bacterial adherence. A total of 77 clinical isolates of Escherichia coli were tested. Cranberry juice inhibited adherence by 75 per cent or more in over 60 per cent of the clinical isolates. Cranberry cocktail was also given to mice in the place of their normal water supply for a period of 14 days. Urine collected from these mice inhibited adherence of E. coli to uroepithelial cells by approximately 80 per cent. Antiadherence activity could also be detected in human urine. Fifteen of 22 subjects showed significant antiadherence activity in the urine 1 to 3 hours after drinking 15 ounces of cranberry cocktail. It is concluded that the reported benefits derived from the use of cranberry juice may be related to its ability to inhibit bacterial adherence.